



Geometric Latent Diffusion Models for 3D Molecule Generation

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Introduction

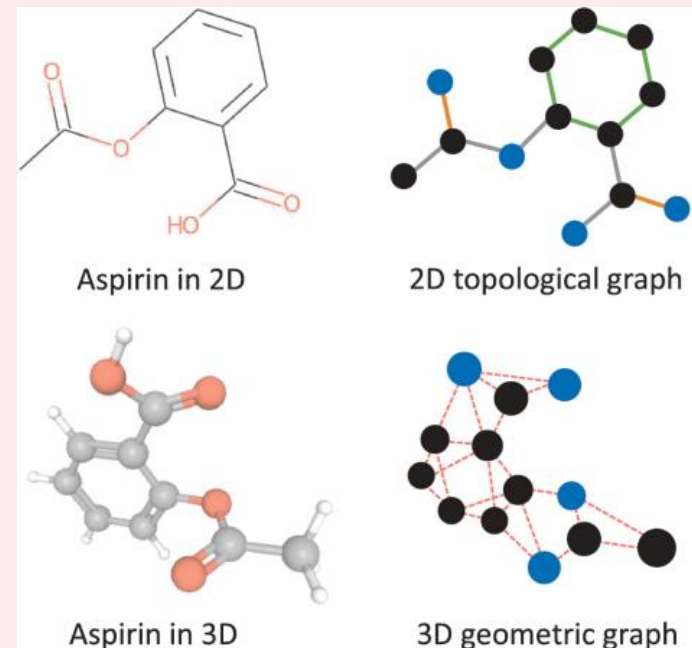
3D molecule generation

1] 분자의 computational 표현

1D : 원자, 결합, 방향족, 가지 를 표시하여 1차원의 string 서열의 형태로 표현 (SMILES)

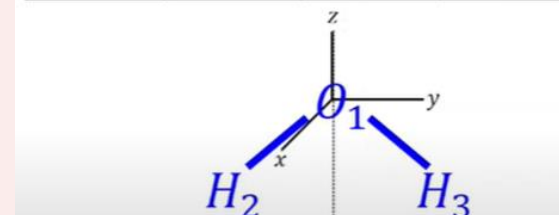
2D : 원자와 결합의 종류, 특성으로 이루어진 그래프 형태로 표현

3D : 공간 상의 point cloud 로 직교 좌표계의 형태로 표현

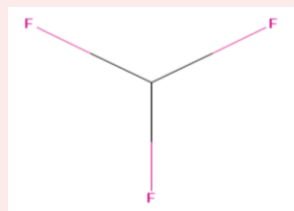
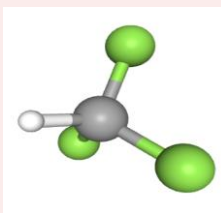


Graph Neural Networks for Molecules

	x	y	z
O_1	0	0	0
H_2	0	-0.79	-0.58
H_3	0	-0.79	-0.58

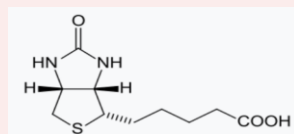
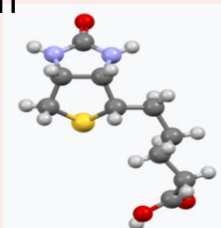


Trifluoromethane



C(F)(F)F

Biotin



Hydrogen bond acceptors	5
Hydrogen bond donors	3
Rotatable bonds	5
Topological polar surface area	103.73
Molecular weight	244.09
XLogP	0.46
No. Lipinski's rules broken	0

OC(=O)CCCCC1SCC2C1NC(=O)N2

Wikipedia

Building molecules “by hand” using Z matrices

Introduction

● 3D molecule generation

2] 분자의 3차원 구조

: 3차원 상의 분자의 탄소-탄소 간 단일 결합에서 회전(rotation), 평행이동(translation), 반사(reflection) 가 일어남에 따라 분자가 변화하지 않음.

- Equivariant transformation

: 분자 내 원자의 3차원 상에서 translation, reflection, rotation 에 따라 transformation property 가 정해짐

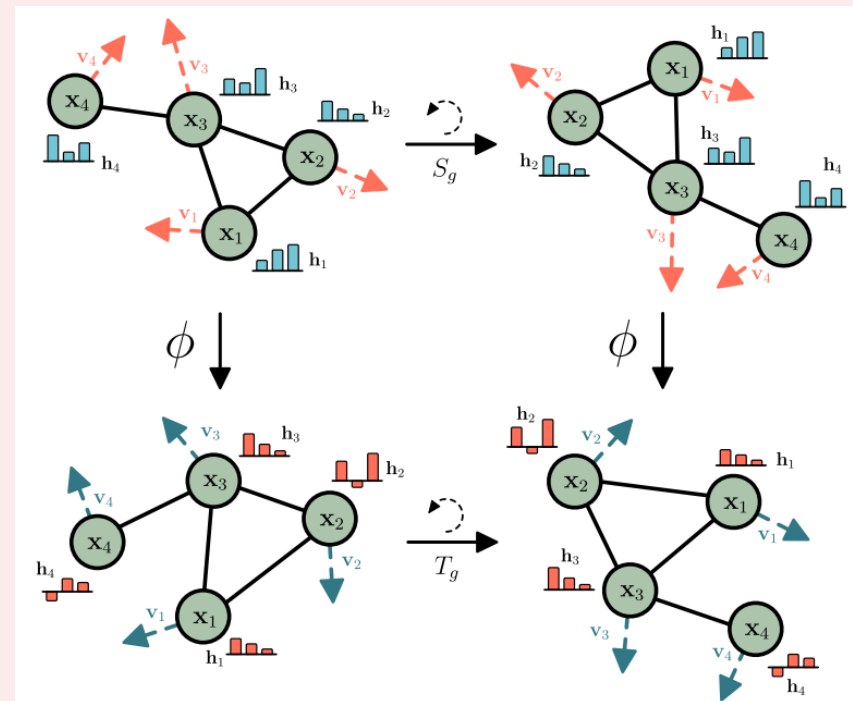


Figure 1. Example of rotation equivariance on a graph with a graph neural network ϕ

Introduction

- 기존 3D molecule generators

1] data-driven methods

: autoregressive model : 원자/분자의 sequence (Gebauer et al. (2019); Luo & Ji (2021))

: flow-based model (Satorras et al. (2021a))

- 복잡한 action space/ action ordering 설계, atomic density grid를 통한 1-step 생성
- > low chemical validity, small molecule size, insufficient model capacity models

2] diffusion-based models

: atomic feature space 이용 (원자 종류, charge, coordinates), target drug 생성 task 학습.

- > Suboptimal due to multimodality, require sophisticated modeling, high dimensionality (Hoogeboom et al., 2022; Wu et al., 2022, Anand & Achim, 2022)

Introduction

- In this paper

Geometric Latent Diffusion Models (GEOLDM) 모델

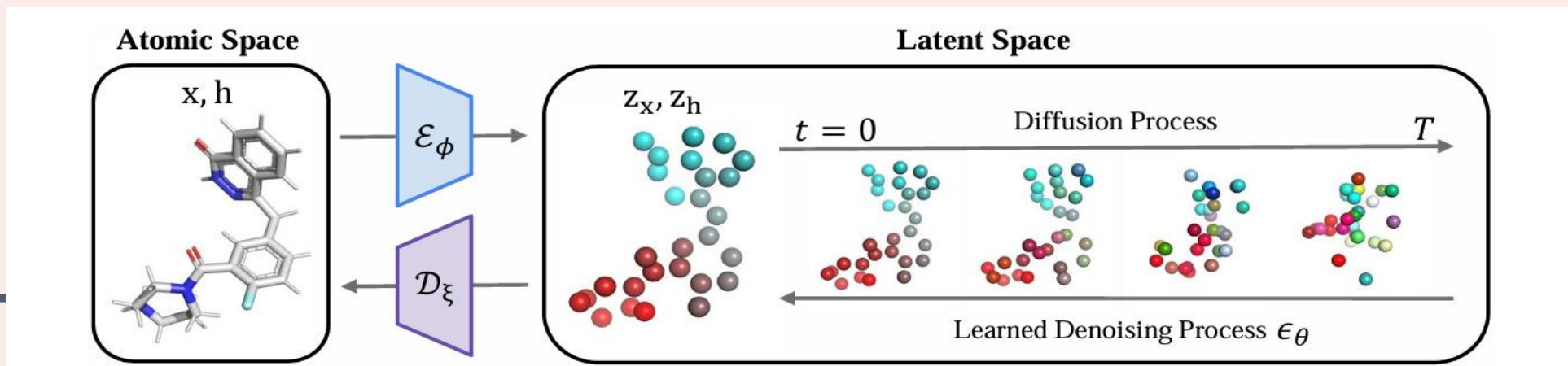
- key point

- 1) roto-translational equivariance constraint 반영 생성 모델

: invariant scalars and equivariant variable 이용

- 2) latent space 이용 diffusion 모델

: continuous latent 생성 autoencoder + latent space 에서 denoising diffusion 모델



Background

- Latent Diffusion model

1) Non-geometric Domains

- Forward process

$$q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

- Reverse process

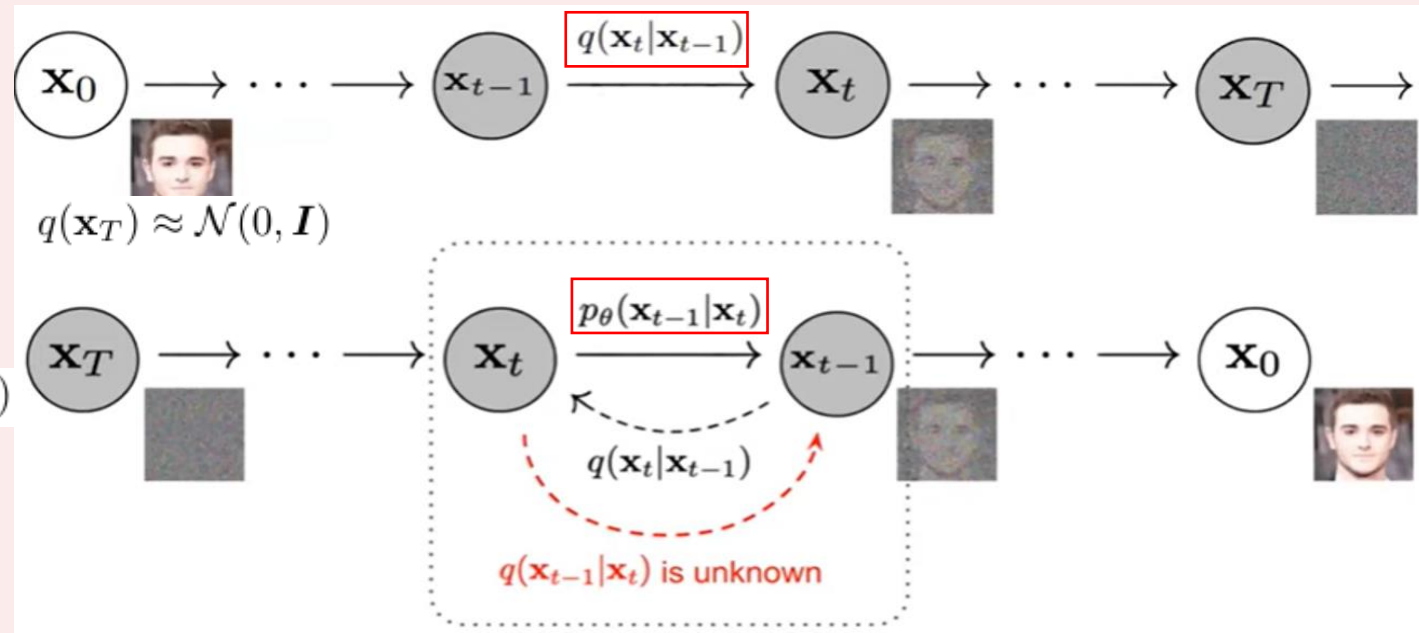
$$p_{\theta}(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

- Gaussian noise $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}),$$

- Denoising

$$p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \rho_t^2 \mathbf{I}),$$



Background

- Latent Diffusion Model

3. Variational lower bound

$$\mathcal{L}_{vlb} = \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0, \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]$$

$$\mathcal{L}_{DM} = \mathbb{E}_{\mathbf{x}_0, \epsilon \sim \mathcal{N}(0, \mathbf{I}), t} \left[w(t) \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2 \right] \quad \text{Song \& Ermon (2019); Ho et al. (2020)}$$

- sampling
$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\alpha_t^2}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \rho_t \epsilon$$

- Problem Definition

3차원 공간에서의 분자 $g = \langle \mathbf{x}, \mathbf{h} \rangle$ 를 atom coordinates matrix $\mathbf{x}$, node feature matrix $\mathbf{h}$ 로 정의

$$\mathcal{G} = \langle \mathbf{x}, \mathbf{h} \rangle, \text{ where } \mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \mathbb{R}^{N \times 3} \quad \mathbf{h} = (\mathbf{h}_1, \dots, \mathbf{h}_N) \in \mathbb{R}^{N \times d}$$

Background

- Equivariance

F 함수가 equivariant 하면, 다음을 만족

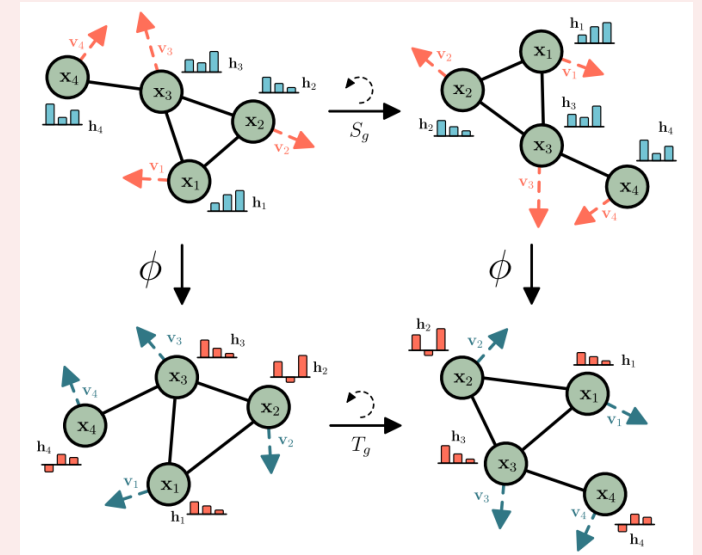
$$\mathcal{F} \circ S_g(\mathbf{x}) = T_g \circ \mathcal{F}(\mathbf{x})$$

S, T : group element g 에 대한 transformation

-> **Special Euclidean group SE(3) Equivariance** 이용

translations, rotations, reflection 에 대한 분자의 equivariance 반영

: invariant scalar 와 equivariant tensor 를 이용



Method

- Geometric Latent Diffusion Models (GEOLDM)

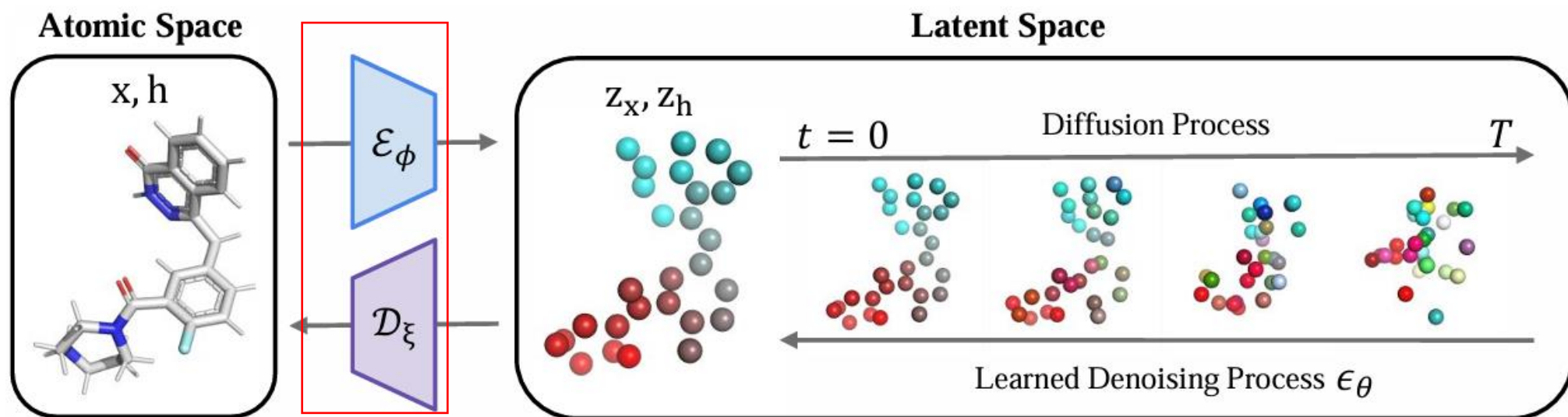


Figure 1. Illustration of GEOLDM. The encoder $\mathcal{E}_\phi$ encodes molecular features $\mathbf{x}, \mathbf{h}$ into equivariant latent variables $\mathbf{z}_x, \mathbf{z}_h$, and the latent diffusion transitions $q(\mathbf{z}_{x,t}, \mathbf{z}_{h,t} | \mathbf{z}_{x,t-1}, \mathbf{z}_{h,t-1})$ gradually added noise until the latent codes converge to Gaussians. Symmetrically, for generation, an initial latent $\mathbf{z}_{x,T}, \mathbf{z}_{h,T}$ is sampled from standard normal distributions and progressively refined by equivariant denoising dynamics $\epsilon_\theta(\mathbf{z}_x, \mathbf{z}_h)$. The final latents $\mathbf{z}_x, \mathbf{z}_h$ are further decoded back to molecular point clouds with the decoder $\mathcal{D}_\xi$.

Method

- Geometric Latent Diffusion Models (GEOLDM)

분자의 3차원 표현 $\mathcal{G} = \langle \mathbf{x}, \mathbf{h} \rangle$

= **Atom coordinates matrix**

$$\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \mathbb{R}^{N \times 3}$$

+

node feature matrix (scalar) - invariant

$$\mathbf{h} = (\mathbf{h}_1, \dots, \mathbf{h}_N) \in \mathbb{R}^{N \times d}$$

1. Geometric Autoencoding

1) Special Euclidean group SE(3) Equivariance 이용 (translations, rotations, reflection)

2) Invariant latent features $\mathbf{z}_x$ (k 개)

$$\mathbf{z} = \langle \mathbf{z}_x, \mathbf{z}_h \rangle \in \mathbb{R}^{N \times (3+k)}$$

Rotation $\mathbf{R}$, translation $\mathbf{h}$ 에 대해 latent space $\mathbf{z}$ 를 학습

$$\mathbf{R}\mathbf{z}_x + \mathbf{t}, \mathbf{z}_h = \mathcal{E}_\phi(\mathbf{R}\mathbf{x} + \mathbf{t}, \mathbf{h}); \mathbf{R}\mathbf{x} + \mathbf{t}, \mathbf{h} = \mathcal{D}_\xi(\mathbf{R}\mathbf{z}_x + \mathbf{t}, \mathbf{z}_h), \quad (5)$$

Method

- Geometric Latent Diffusion Models (GEOLDM)

2. Geometric latent diffusion model

$$\mathbf{z} = \mathcal{E}_\phi(\mathbf{x}, \mathbf{h})$$

Equivalant 하기 때문에 distribution 에 대한 학습이 가능

$$p_\theta(\mathbf{z}_{\mathbf{x},t-1}, \mathbf{z}_{\mathbf{h},t-1} | \mathbf{z}_{\mathbf{x},t}, \mathbf{z}_{\mathbf{h},t}) = \\ p_\theta(\mathbf{R}\mathbf{z}_{\mathbf{x},t-1}, \mathbf{z}_{\mathbf{h},t-1} | \mathbf{R}\mathbf{z}_{\mathbf{x},t}, \mathbf{z}_{\mathbf{h},t}), \quad \forall \mathbf{R}.$$

$$\mathcal{L}_{LDM} = \mathbb{E}_{\mathcal{E}(\mathcal{G}), \epsilon \sim \mathcal{N}(0, \mathbf{I}), t} [w(t) ||\epsilon - \epsilon_\theta(\mathbf{z}_{\mathbf{x},t}, \mathbf{z}_{\mathbf{h},t}, t)||^2],$$

3. Conditional generation $\epsilon_\theta(\mathbf{z}, t, s)$

feature 에 concatenate 하는 방법을 사용

Method

Algorithm 1 Training Algorithm of GEOLDM

```

1: Input: geometric data  $\mathcal{G} = \langle \mathbf{x}, \mathbf{h} \rangle$ 
2: Initial: encoder network  $\mathcal{E}_\phi$ , decoder network  $\mathcal{D}_\xi$ , de-
   noising network  $\epsilon_\theta$ 
3: First Stage: Autoencoder Training
4: while  $\phi, \xi$  have not converged do
5:    $\mu_x, \mu_h \leftarrow \mathcal{E}_\phi(\mathbf{x}, \mathbf{h})$  {Encoding}
6:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
7:   Subtract center of gravity from  $\epsilon_x$  in  $\epsilon = [\epsilon_x, \epsilon_h]$ 
8:    $\mathbf{z}_x, \mathbf{z}_h \leftarrow \epsilon \odot \sigma_0 + \mu$  {Reparameterization}
9:    $\tilde{\mathbf{x}}, \tilde{\mathbf{h}} \leftarrow \mathcal{D}_\xi(\mathbf{z}_x, \mathbf{z}_h)$  {Decoding}
10:   $\mathcal{L}_{AE} = \text{reconstruction}([\tilde{\mathbf{x}}, \tilde{\mathbf{h}}], [\mathbf{x}, \mathbf{h}]) + \mathcal{L}_{reg}$ 
11:   $\phi, \xi \leftarrow \text{optimizer}(\mathcal{L}_{AE}; \phi, \xi)$ 
12: end while
13: Second Stage: Latent Diffusion Models Training
14: Fix encoder parameters  $\phi$ 
15: while  $\theta$  have not converged do
16:   $\mathbf{z}_{x,0}, \mathbf{z}_{h,0} \sim q_\phi(\mathbf{z}_x, \mathbf{z}_h | \mathbf{x}, \mathbf{h})$  {As lines 5-8}
17:   $t \sim \mathbf{U}(0, T), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
18:  Subtract center of gravity from  $\epsilon_x$  in  $\epsilon = [\epsilon_x, \epsilon_h]$ 
19:   $\mathbf{z}_{x,t}, \mathbf{z}_{h,t} = \alpha_t[\mathbf{z}_{x,0}, \mathbf{z}_{h,0}] + \sigma_t \epsilon$ 
20:   $\mathcal{L}_{LDM} = ||\epsilon - \epsilon_\theta(\mathbf{z}_{x,t}, \mathbf{z}_{h,t}, t)||^2$ 
21:   $\theta \leftarrow \text{optimizer}(\mathcal{L}_{LDM}; \theta)$ 
22: end while
23: return  $\mathcal{E}_\phi, \mathcal{D}_\xi, \epsilon_\theta$ 

```

Algorithm

$$\mathcal{L}_{AE} = \mathcal{L}_{recon} + \mathcal{L}_{reg},$$

$$\mathcal{L}_{recon} = -\mathbb{E}_{q_\phi(\mathbf{z}_x, \mathbf{z}_h | \mathbf{x}, \mathbf{h})} p_\xi(\mathbf{x}, \mathbf{h} | \mathbf{z}_x, \mathbf{z}_h),$$

$$\mathcal{L}_{LDM} = \mathbb{E}_{\mathcal{E}(\mathcal{G}), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), t} [w(t) ||\epsilon - \epsilon_\theta(\mathbf{z}_{x,t}, \mathbf{z}_{h,t}, t)||^2],$$

Algorithm 2 Sampling Algorithm of GEOLDM

```

1: Input: decoder network  $\mathcal{D}_\xi$ , denoising network  $\epsilon_\theta$ 
2:  $\mathbf{z}_{x,T}, \mathbf{z}_{h,T} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
3: for  $t$  in  $T, T-1, \dots, 1$  do
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  {Latent Denoising Loop}
5:   Subtract center of gravity from  $\epsilon_x$  in  $\epsilon = [\epsilon_x, \epsilon_h]$ 
6:    $\mathbf{z}_{t-1} = \frac{1}{\sqrt{1-\beta_t}}(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t^2}}\epsilon_\theta(\mathbf{z}_t, t)) + \rho_t \epsilon$ 
7: end for
8:  $\mathbf{x}, \mathbf{h} \sim p_\xi(\mathbf{x}, \mathbf{h} | \mathbf{z}_{x,0}, \mathbf{z}_{h,0})$  {Decoding}
9: return  $\mathbf{x}, \mathbf{h}$ 

```

Result

- **Experiment**

Dataset : QM9, GEOM-DRUG

CONTENTS

1. Molecular Modeling and Generation

: chemically valid, structurally diverse molecules

- 원자 간 종류와 거리를 통해 결합 종류 예측(single, double, triple, none)

Validity, uniqueness : RDKIT 이용

2. Controllable Molecule Generation

: generating target molecules with desired chemical properties

- polarizability, orbital energies와 gap, dipole moment, heat property 조건 생성



Result

1. Molecular Modeling and Generation

10,000개의 샘플 생성 후 비교

Atom Stability, Molecular stability, Validity, Validity & Uniqueness

# Metrics	QM9				DRUG	
	Atom Sta (%)	Mol Sta (%)	Valid (%)	Valid & Unique (%)	Atom Sta (%)	Valid (%)
Data	99.0	95.2	97.7	97.7	86.5	99.9
ENF	85.0	4.9	40.2	39.4	-	-
G-Schnet	95.7	68.1	85.5	80.3	-	-
GDM	97.0	63.2	-	-	75.0	90.8
GDM-AUG	97.6	71.6	90.4	89.5	77.7	91.8
EDM	98.7	82.0	91.9	90.7	81.3	92.6
EDM-Bridge	98.8	84.6	92.0*	90.7	82.4	92.8*
GRAPHLDM	97.2	70.5	83.6	82.7	76.2	97.2
GRAPHLDM-AUG	97.9	78.7	90.5	89.5	79.6	98.0
GEO LDM	98.9 ± 0.1	89.4 ± 0.5	93.8 ± 0.4	92.7 ± 0.5	84.4	99.3

Result

2. Controllable Molecule Generation

- Molecular property prediction 을 통해 각 property 조건에서의 생성 성능 MAE 확인

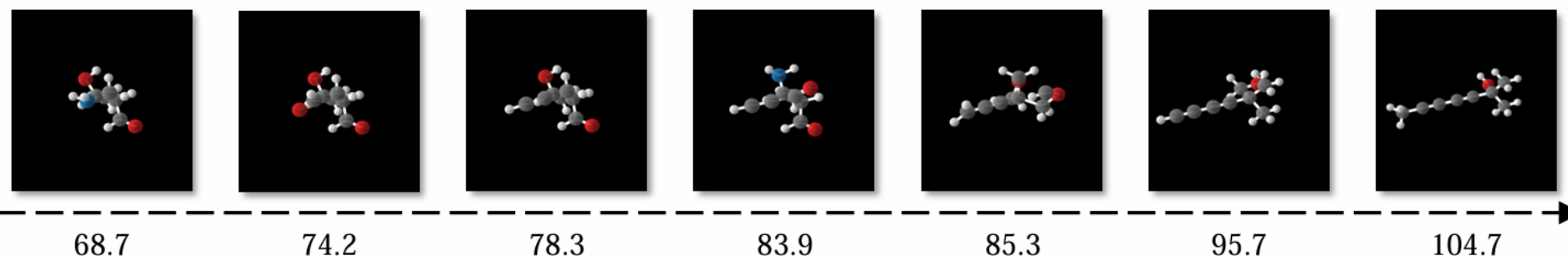


Figure 3. Molecules generated by conditional GEO-LDM. We conduct controllable generation with interpolation among different Polarizability α values with the same reparametrization noise ϵ . The given α values are provided at the bottom.

Table 2. Mean Absolute Error for molecular property prediction. A lower number indicates a better controllable generation result. Results are predicted by a pretrained EGNN classifier ω on molecular samples extracted from individual methods.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	μ	C_v
Units	Bohr ³	meV	meV	meV	D	$\frac{\text{cal}}{\text{mol}} \text{K}$
QM9*	0.10	64	39	36	0.043	0.040
Random*	9.01	1470	645	1457	1.616	6.857
N_{atoms}	3.86	866	426	813	1.053	1.971
EDM	2.76	655	356	584	1.111	1.101
GEO-LDM	2.37	587	340	522	1.108	1.025

Conclusion

- **Roto-translational equivariance constraint 적용 – simple**
 - **Latent Space diffusion 모델 사용**
 - 1) regularized : smoother distribution 학습 가능
 - 2) structures' likelihood 를 모델 : more expressive 생성
 - 3) low dimensionality : complexity 감소
 - 4) better control over generation process
 - 5) molecular problem application 확장 : drug design, antigen specific antibody 생성
-



Thank you

